

WQBEL BIOLOGIST RECOMMENDATIONS
FACT/DECISION SHEET
Cadillac WWTP - MI0020257 - REISSUANCE
PREPARED BY: GLEN SCHMITT
October 17, 2022

Background

The City of Cadillac submitted a National Pollutant Discharge Elimination System (NPDES) permit application on March 25, 2022, for reissuance of the Cadillac WWTP facility located at 1121 Plett Road, Cadillac, Michigan, 49601, Wexford County. The current NPDES permit was issued on August 30, 2019, effective on September 1, 2019, and expired on October 1, 2022, but has been extended.

The current NPDES permit authorizes to continuously discharge 3.2 million gallons per day (MGD) (this flow is not to be considered a limitation or actual capacity) of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River at Latitude 44.2657, Longitude -85.3985.

The Clam River has a lowest 95 percent exceedance, harmonic mean, and 90dQ10 low flows of 2.6, 13, and 3.4 cubic feet per second (cfs), respectively (Low Flow File No. 10194 [Wexford County]). The Clam River is designated as warmwater within this reach and is not protected as a public water supply source.

A total hardness value of 70 milligrams per liter (mg/l) was used for this review and was collected on August 19, 2021, upstream of Outfall 001 at Latitude 44.263376, Longitude -85.4011 (STORET Number 830054) by Permits Section staff.

Sources

Sources of information used for this review to develop the water quality-based effluent limits (WQBELs) include the facility's NPDES permit reissuance application, the current NPDES permit, the Permits Section Toxics facility file, discharge monitoring report data from November 2017 through August 2022, a Point Source Monitoring Survey conducted on May 9, 2018, and United States Geological Survey topographical maps. Based upon our review of this information, we have the following recommendations:

Designated Use Attainment Status

- 1) The 2022 Water Quality and Pollution Control in Michigan Sections 303(d), 305(b), and 314 Integrated Report indicates the Clam River (Assessment Unit ID (AUID) 040601020303-01) is Not Supporting Michigan Water Quality Standards (WQS) for the Fish Consumption, Partial Body Contact Recreation, and Total Body Contact Recreation designated uses. The Fish Consumption designated use is not supporting Michigan WQS due to polychlorinated biphenyls (PCBs). The Partial Body Contact Recreation and Total Body Contact Recreation designated uses are not supporting Michigan WQS due to *Escherichia coli* (*E. coli*). The designated uses for Agriculture, Coldwater Fishery, Industrial Water Supply, Navigation, and Other Indigenous Aquatic Life and Wildlife are listed as fully supporting. The designated use for Warm Water Fishery is listed as not assessed. A statewide Total Maximum Daily Load (TMDL) for PCBs was approved by the United States Environmental Protection Agency (USEPA) on September 26, 2017. A statewide TMDL for *E. coli* was approved by USEPA on August 31, 2020.
- 2) A search of available information for endangered, threatened, or sensitive species did not identify any aquatic species in the Clam River near Outfall 001 as being endangered, threatened, or sensitive species. The following Townships and Ranges were evaluated: Town 22N, Range 09W, Section 33.

Recommendations for Outfall 001

- 3) The current NPDES permit requires daily flow reporting. The Outfall 001 final effluent flow from November 2017 through August 2022 ranged from 1.09 to 2.97 MGD, averaging 1.76 MGD as compared to the design flow of 3.2 MGD. Based on this information, we recommend the draft NPDES permit retain the daily monitoring requirement for the Outfall 001 final effluent flow.
- 4) The current NPDES permit includes daily minimum and daily maximum final effluent limitations for pH of 6.5 and 9.0 Standard Units (S.U.), respectively. Compliance with these limits is collected as five times weekly grab samples. Levels of pH in the Outfall 001 final effluent from November 2017 through

August 2022 ranged from 6.2 to 8.4 S.U., averaging 7.2 S.U. There were two days during this period when the daily minimum limit of 6.5 S.U. was exceeded. Based on this information, we recommend the draft NPDES permit retain the current final effluent limits and monitoring requirements for pH.

- 5) The current NPDES permit includes a monthly average total phosphorus concentration limit of 0.5 milligram per liter (mg/l) and a loading limit of 13 pounds per day (lbs./day) from May through March and 0.65 mg/l (17 lbs./day) for April. Compliance with these limitations is collected five times weekly as 24-hour composite samples. Concentration and loading of total phosphorus in the Outfall 001 final effluent from November 2017 through August 2022 ranged from 0.05 mg/l to 52.54 mg/l, averaging 0.42 mg/l as a concentration and 0.11 lbs./day to 33.7 lbs./day, averaging 2.9 lbs./day as a loading. The monthly average concentration limit was exceeded in May 2022. Based on this information, we recommend the draft NPDES permit retain the final effluent limitations and monitoring requirements for total phosphorus.
 - a. The load limitation of 13 lbs./day is based on the design flow of 3.2 MGD, the limit of 0.5 mg/l, and 8.34 pounds (the weight of one gallon of water). $3.2 \text{ MGD} \times 0.5 \text{ mg/l} \times 8.34 \text{ pounds} = 13 \text{ pounds}$.
 - b. The load limitation of 17 lbs./day is based on the design flow of 3.2 MGD, the limit of 0.65 mg/l, and 8.34 pounds (the weight of one gallon of water). $3.2 \text{ MGD} \times 0.65 \text{ mg/l} \times 8.34 \text{ pounds} = 17 \text{ pounds}$.
 - c. Historically, the Clam River was known to have problems with excessive plant growth due to excessive nutrients from point source discharges. The limitations for total phosphorus have been in place for several permit cycles.
- 6) It is understood that ultraviolet disinfection is used to disinfect the final effluent. There is no information indicating chlorine is used for disinfection. Based on this information, we have no recommendations for total residual chlorine. However, if chlorine is used, we recommend the draft NPDES permit include a daily maximum WQBEL of 38 micrograms per liter (ug/l) with daily compliance monitoring and samples collected as grab samples.
- 7) The current NPDES permit includes a monthly average and daily maximum WQBEL for total copper of 11 micrograms per liter (ug/l) (0.3 lbs./day) and 31 ug/l (0.82 lbs./day), respectively. Compliance with these limitations is conducted as weekly 24-hour composite samples. Concentrations of total copper in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.075 ug/l to 10 ug/l) to 24 ug/l. Total copper data collected from November 2017 through August 2019 was not used as the quantification level report (<50 ug/l) is not considered sufficiently sensitive. These data indicate there is not a reasonable potential for total copper to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan Water Quality Standards (WQS). We recommend the draft NPDES permit remove the limitations for total copper. However, due to concerns with elevated concentrations of total copper, we recommend the draft NPDES permit include a weekly monitoring requirement with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Due to the quantification levels reported during the period reviewed, we do not recommend monitoring frequency reduction language to be included in the draft NPDES permit.
 - a. Weekly monitoring is recommended due to elevated concentrations reported and due to the quantification levels reported. This monitoring frequency will allow for a robust dataset to be evaluated at the next reissuance.
- 8) The current NPDES permit includes quarterly monitoring requirement for total nickel with samples collected as 24-hour composite samples. Concentrations of total nickel in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.075 ug/l to 5.0 ug/l) to 7.8 ug/l. These data indicate there is not a reasonable potential for total nickel to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include monitoring for total nickel in the Additional Monitoring Requirements.

- 9) The current NPDES permit includes a monthly average WQBEL for total selenium of 5.7 ug/l (0.15 lbs./day). Compliance with this limitation is conducted as monthly 24-hour composite samples. Concentrations of total selenium in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.081 ug/l to 5.0 ug/l) to 2.0 ug/l. These data indicate there is not a reasonable potential for total selenium to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit remove the final effluent limitations for total selenium and include monitoring for total selenium as part of the Additional Monitoring Requirements.
- 10) The current NPDES permit includes a monthly average WQBEL for Bis(2-ethylhexyl) phthalate of 31 ug/l (0.82 lbs./day). Compliance with this limitation is conducted as monthly 24-hour composite samples. Concentrations of Bis(2-ethylhexyl) phthalate in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 3.9 ug/l to 4.7 ug/l) to 4.1 ug/l. These data indicate there is not a reasonable potential for Bis(2-ethylhexyl) phthalate to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit remove the final effluent limitations for Bis(2-ethylhexyl) phthalate and include monitoring for Bis(2-ethylhexyl) phthalate as part of the Additional Monitoring Requirements.
- 11) The current NPDES permit includes a Level Currently Achievable (LCA) Total Mercury limitation of 2.0 nanograms per liter (ng/l) and a loading limitation of 0.000053 lbs./day calculated as a 12-month rolling average. Compliance with this limitation is collected as quarterly grab samples. Part I.A.4 of the current NPDES permit includes requirements for a Pollutant Minimization Program for Total Mercury. Concentrations of total mercury in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level <0.5 ng/l) to 2.82 ng/l. Based on the final effluent data for total mercury, there is a reasonable potential for total mercury to be discharged from Outfall 001 at levels exceeding Michigan WQS. We recommend the draft NPDES permit retain the LCA of 2.0 ng/l (0.000053 lbs./day). Compliance with this limitation shall be collected as quarterly grab samples using USEPA method 1669 and analyzed using USEPA method 1631 with a quantification level of 0.5 ng/l. In addition, we recommend the PMP language in Part I.A.4 be retained in the draft permit. We do not recommend the draft NPDES permit include monitoring frequency reduction language for total mercury.
- a. Review of total mercury final effluent data and reasonable potential analysis was consistent with Water Resources Division, Policy and Procedure, WRD-004, Calculation of Level Currently Achievable (LCA) for Mercury in Proposed National Pollutant Discharge Elimination System (NPDES) Permits.
 - i. The “n” value in the PEL Sheet was put at 4 based on a quarterly monitoring frequency.
 - b. Quarterly monitoring is recommended as the facility has consistently demonstrated low concentrations of total mercury and compliance with the current LCA.
 - c. A monitoring frequency reduction is not recommended as a sufficient dataset is necessary during future reviews.
- 12) The current NPDES permit requires Whole Effluent Toxicity (WET) testing for acute and chronic toxicity as part of the conditions in Part I.A.3, Additional Monitoring Requirements with testing conducted using Fathead minnow and *Ceriodaphnia dubia* (*C. dubia*) as the test species for a total of eight tests (four tests on each species). Acute toxicity test results from November 2017 through August 2022 for *C. dubia* and Fathead minnow never measured any acute toxicity (all results were 0.0 acute toxic units (TUa)). Chronic toxicity for *C. dubia* ranged from 0.0 chronic toxic units (TUc) to 5.6 TUc with toxicity occurring during the two tests conducted in August 2019. No other tests demonstrated chronic toxicity. Only one of five tests were chronically toxic to Fathead minnow with a result of 1.1 TUc.

Based on this information, there is a reasonable potential for the Outfall 001 final effluent to be chronically toxic to aquatic life. We recommend the draft NPDES permit include a chronic and acute WET limitation of 1.1 TUc and 1.0 TUa, respectively. Compliance with these limitations shall be quarterly with samples collected as 24-hour composite samples and testing using Fathead minnow and *C. dubia* as the test species. Testing shall be in accordance with chronic toxicity testing methods.

Appropriate language shall also be included in the draft NPDES permit to allow the permittee to request a monitoring frequency reduction to no less than annually after two years of quarterly testing has been completed based on the effective date of the NPDES permit.

- 13) Arsenic monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of arsenic in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level of <1.0 ug/l) to 76 ug/l. These data indicate there is a reasonable potential for arsenic to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 16 ug/l (0.42 lbs./day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 14) Beryllium monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of beryllium in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.4 ug/l to <1.0 ug/l) to 8.1 ug/l. These data indicate there is a reasonable potential for beryllium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 4.1 ug/l (0.11 lbs./day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 15) Cadmium monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of cadmium in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level <0.2 ug/l) to 9.5 ug/l. These data indicate there is a reasonable potential for cadmium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL and daily maximum WQBEL of 4.1 ug/l (0.11 lbs./day) and 12 ug/l (0.32 lbs./day), respectively. Compliance with these limitations shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 0.2 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 16) Silver monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of silver in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.2 ug/l to <0.5 ug/l) to 1.8 ug/l. These data indicate there is a reasonable potential for silver to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 1.4 ug/l (0.036 lbs./day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 0.5 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 17) Available cyanide monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of available cyanide in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 1.0 ug/l to 2.0 ug/l) to 8.4 ug/l. These data indicate there is a reasonable potential for

available cyanide to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 5.9 ug/l (0.16 lbs./day). Compliance with this limitation shall be monthly with samples collected as a grab sample and analyzed using Method OIA-1677 with a quantification level of 2.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than annually after two years of monthly monitoring based on the effective date of the NPDES permit.

- 18) Benzidine monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of benzidine in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level of <0.1 ug/l) to 1.8 ug/l. These data indicate there is a reasonable potential for benzidine to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 0.12 ug/l (0.003 lbs./day). Compliance with this limitation shall be weekly with samples collected as a grab sample and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 0.1 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than annually after two years of weekly monitoring have been completed based on the effective date of the NPDES permit.
- 19) Bromomethane monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of bromomethane in the Outfall 001 final effluent from November 2017 through August 2022 were all reported as not detected with quantification levels of <1.0 ug/l for two results and <5.0 ug/l for the other two results. The samples that reported a quantification level of <5.0 ug/l were reported with a quantification level that is not considered sufficiently sensitive when compared to the theoretical WQBELs. Based on this information, we recommend the draft NPDES permit include a quarterly monitoring requirement for bromomethane. Samples for bromomethane shall be collected as grab samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than annually after two years of quarterly monitoring have been completed based on the effective date of the NPDES permit.
- 20) The current NPDES permit includes 2X annual monitoring for Perfluorooctane sulfonate (PFOS) with samples collected as grab samples. Concentrations of PFOS in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from <1.6 ng/l to <1.9 ng/l) to 7.8 ng/l. These data indicate there is a reasonable potential for PFOS to be discharged from Outfall 001 at concentrations exceeding Michigan WQS. We recommend a monthly average WQBEL of 0.012 ug/l (0.00032 lbs./day). Compliance with this limitation shall be quarterly with samples collected as grab samples. Analysis for PFOS shall be with method ASTM D7979 or an isotope dilution method (sometimes referred to as Method 537 modified) at a quantification level of 2.0 ng/l. Language shall also be included to allow for a monitoring frequency reduction request to no less than annually to be submitted after 10 equally spaced samples over a minimum of three months have been collected.
- 21) The current NPDES permit includes 2X annual monitoring for Perfluorooctanoic acid (PFOA) with samples collected as grab samples. Concentrations of PFOA in the Outfall 001 final effluent from November 2017 through August 2022 ranged from 1.9 ng/l to 20 ng/l. These data indicate there is not a reasonable potential for PFOA to be discharged from Outfall 001 at concentrations exceeding Michigan WQS. We have no recommendations for PFOA.
 - a. The theoretical WQBELs for PFOA are 0.28 ug/l (0.0075 lbs./day) as a Monthly Average WQBEL and 15,000 ug/l (400 lbs./day) as a Daily Maximum WQBEL.
- 22) The current NPDES permit does not include any final effluent limitations or monitoring requirements for Perfluorobutanesulfonate (PFBS). Final effluent data collected from the Outfall 001 final effluent from November 2017 through August 2022 ranged from 3.1 ng/l to 58 ng/l. These data indicate there is not

a reasonable potential for PFBS to be discharged from Outfall 001 at concentrations exceeding Michigan WQS. We have no recommendations for PFBS.

- a. The theoretical WQBELs for PFBS are 1,100 ug/l (30 lbs./day) as a Monthly Average WQBEL and 240,000 ug/l (6,400 lbs./day) as a Daily Maximum WQBEL.

- 23) As part of the Chloride and Sulfate Implementation Plan that became effective in February 2021, we recommend the draft NPDES permit include a monthly monitoring requirement for Chloride and Sulfate for the full term of the permit. Samples for Chloride and Sulfate shall be collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1,000 ug/l for Chloride and 2,000 ug/l for Sulfate. We do not recommend the draft NPDES permit include a monitoring frequency reduction as a sufficient dataset will be required for review at the next reissuance.

Other Recommendations

- 24) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.
- 25) All other data reviewed indicated there was not a reasonable potential for these parameters to be discharged at concentrations exceeding Michigan WQS. These data were either provided with the NPDES permit application, in the Permits Section toxics facility file, or in MiWaters. Parameters with results reported as all Not Detected were also reviewed. Based on this information, we have no recommendations for these parameters at this time. Refer to the PEL sheet and permit application for a list of other parameters reviewed.

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Permits Section, WRD, EGLE
Date:	10/17/2022

Outf.	Authorized Flow		Disch. Type *	Receiving Water	Lake?	FLOWS (cfs)			pH Hardness	
	mgd	cfs				95% Exc.	H. Mean	90Q10		
001	3.2	4.95		Clam River	☐	2.6	13.0	3.4	7	70

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

TYPE OF REVIEW: NPDES Permit Reissuance

LOW FLOW FILE NO.: 10194 (Wexford County)

HARDNESS SOURCE: STORET 830054 collected on August 19, 2021 by Permits Section Staff.

PEL SHEET VERSION: 7-22-2022

Review the Following Information Sources

☐ **Permit Application**

☐ **Facility/VGW File**

☐ **MiWaters Documents & Files**

☐ **Current Permit**

☐ **WTA's**

☐ **Staff Report #'s**

☐ **WET Tests**

☐ **Other**

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	70 mg/L CaCO3
Outfall:	001	pH:	7
Date:	10/17/2022	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met. Translator	Background Conc.
		FCV	vd	HNV	vd	HCV	vd	WV	vd	FAV	vd		
Copper	7440508	6.60295443	1199707	38000	1200512	NA	0	NA	0	19.20666594	1199707	1.5	4
Nickel	7440020	38.46014329	1199707	210000	1199706	NA	0	NA	0	692.5442623	1199707	1.1	3
Selenium	7782492	5	1199707	2700	1199704	NA	0	NA	0	120	1199808	1	0
Bis(2-ethylhexyl)phthalate #	117817	ID*	199809	160	1199711	18	1201402	NA	0	285	2199809	1	0
Mercury @	7439976	0.77	1199707	0.0018	1199707	NA	0	0.0013	1199707	2.8	1199707	1	**
Perfluorooctane sulfonate @	1763231	140	2201408	0.012	1201403	NA	0	NA	0	1600	2201408	1	0
Perfluorooctanoic acid	335671	880	2201007	0.17	1202012	NA	0	NA	0	15000	2201007	1	0
Perfluorobutanesulfonate	375735	24000	1201112	670	1202112	NA	0	NA	0	240000	1201112	1	0
Cyanide, free	57125	5.2	1199707	48000	1199707	NA	0	NA	0	44	1199707	1	0
Benzidine #	92875	2.7	2200902	3700	1201003	0.073	1201003	NA	0	49	2200902	1	0
3,3'-Dichlorobenzidine #	91941	4.5	2199710	950	2199709	0.2	2199709	NA	0	81	2199710	1	0
Hexachlorobenzene # @	118741	ID*	199801	0.046	1199707	0.00045	1199707	0.0003	1199704	1200	3199801	1	0
Hexachlorobutadiene # @	87683	1	2199904	0.098	1199704	0.35	2199704	0.053	1199906	15	2199904	1	0
Hexachlorocyclopentadiene	77474	0.07	2202109	450	1202110	NA	0	NA	0	0.54	2202109	1	0
2-Chlorophenol	95578	18	2200806	400	2200703	NA	0	NA	0	320	2200806	1	0
Bis(2-chloroisopropyl)ether #	108601	540	3199709	47000	1199711	290	1199711	NA	0	9700	3199709	1	0
Bromomethane	74839	4.2	2201401	2600	1199708	NA	0	NA	0	75	2201401	1	0
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

** = Background exceeds standards.

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.6 cfs	Conc.	Load (lbs/day)
Outfall:	001			H. Mean Flow:	13 cfs	Hg Loading Calculated based on LCA (ng/L) of :	2 0.000053376
Date:	10/17/2022			90Q10 Flow:	3.4 cfs	TP Loading based on concentration (mg/L) of :	0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Copper	10.67920379	0.285006591	62928.9381	1679.448	#VALUE!	#VALUE!	#VALUE!	#VALUE!	28.81	0.768881	15.75	15.75	No Reasonable Potential. Remove limits, but retain weekly monitoring due to concerns.	
Nickel	47.46386248	1.266715562	347777.719	9281.492	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples	761.7987	7.6061	14.60855	No Reasonable Potential. Remove limits & monitoring and move to Additional Monitoring Requirements.	
Selenium	5.65609375	0.15094983	4471.45313	119.3341	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples	120	1.490797	5.369074	No Reasonable Potential. Remove limits & monitoring and move to Additional Monitoring Requirements.	
Bis(2-ethylhexyl)phthalate #	#VALUE!	#VALUE!	264.975	7.071653	29.8096875	0.79556094	#VALUE!	#VALUE!	PEQ >= +10 Samples	285	0.311475	1.285597	No Reasonable Potential. Remove limits & monitoring and move to Additional Monitoring Requirements.	
Mercury @	0.77	0.02054976	0.0018	4.8E-05	NA	#VALUE!	0.0013	3.4694E-05	2.8	0.074726	0.003548	0.003548	Reasonable Potential. Retain the LCA of 2.0 ng/l (0.000053 lbs./day), monthly monitoring & PMP.	
Perfluorooctane sulfonate @	140	3.73632	0.012	0.00032	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples	1600	0.001345	0.001977	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 0.012 ug/l (0.00032 lbs./day).	
Perfluorooctanoic acid	995.4725	26.56717008	0.28153594	0.007514	#VALUE!	#VALUE!	#VALUE!	#VALUE!	15000	400.32	0.038	0.038	No Reasonable Potential. No Recommendation.	
Perfluorobutanesulfonate	27149.25	724.559184	1109.58281	29.61255	#VALUE!	#VALUE!	#VALUE!	#VALUE!	240000	6405.12	0.1102	0.1102	No Reasonable Potential. No Recommendation.	
Cyanide, free	5.8823375	0.156987823	79492.5	2121.496	#VALUE!	#VALUE!	#VALUE!	#VALUE!	44	1.174272	10.08	10.08	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 5.9 ug/l (0.16 lbs./day), monthly monitoring.	
Benzidine #	3.054290625	0.081512908	6127.54688	163.532	0.120894844	0.003226442	#VALUE!	#VALUE!	49	1.307712	4.14	4.14	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 0.12 ug/l (0.003 lbs./day), weekly monitoring.	
3,3'-Dichlorobenzidine #	5.090484375	0.135854847	1573.28906	41.98794	0.33121875	0.008839566	#VALUE!	#VALUE!	81	2.161728	0	0	Not Detected. Sufficiently sensitive QL used. No Recommendation.	
Hexachlorobenzene # @	#VALUE!	#VALUE!	0.046	0.001228	0.00045	1.20096E-05	0.0003	8.0064E-06	1200	32.0256	0	0	Not Detected. Sufficiently sensitive QL used. No Recommendation.	
Hexachlorobutadiene # @	1	0.026688	0.098	0.002615	0.35	0.0093408	0.053	0.00141446	15	0.40032	0	0	Not Detected. Sufficiently sensitive QL used. No Recommendation.	
Hexachlorocyclopentadiene	0.079185313	0.002113298	745.242188	19.88902	#VALUE!	#VALUE!	#VALUE!	#VALUE!	0.54	0.014412	0	0	Not Detected. Sufficiently sensitive QL used. No Recommendation.	
2-Chlorophenol	20.3619375	0.543419388	662.4375	17.67913	#VALUE!	#VALUE!	#VALUE!	#VALUE!	320	8.54016	0	0	Not Detected. Sufficiently sensitive QL used. No Recommendation.	
Bis(2-chloroisopropyl)ether #	610.858125	16.30258164	77836.4063	2077.298	480.2671875	12.8173707	#VALUE!	#VALUE!	9700	258.8736	0	0	Not Detected. Sufficiently sensitive QL used. No Recommendation.	
Bromomethane	4.75111875	0.126797857	4305.84375	114.9144	#VALUE!	#VALUE!	#VALUE!	#VALUE!	75	2.0016	0	0	Not Detected. Sufficiently sensitive QL used in 2 of 4 samples. Recommend quarterly monitoring.	
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

* SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY											
Parameter:	Copper	Source	Nickel	Source	Selenium	Source	Bis(2-ethylhexyl)phthalate #	Source	Mercury @	Source	
No. Nondetect	51	DMRs/MiWaters	142	DMRs/MiWaters	156	DMRs/MiWaters		37	DMRs/MiWaters	4	DMRs/MiWaters
Data	7.6	5/9/2018 CSI	4.7	5/9/2018 CSI	0.62	7/2/2020		4.1	6/8/2022	0.00176	11/20/2017
	15	12/11/2019		5	9/4/2019	0.46	7/8/2020			0.000606	5/23/2018
	10	12/26/2019		5.2	10/12/2019	0.55	7/11/2020			0.00079	11/15/2018
	10.7	1/2/2020		3.2	7/2/2020	0.57	7/20/2020			0.00061	2/21/2019
	11.2	1/8/2020		3.1	7/8/2020	0.58	7/28/2020			0.00048	5/21/2019
	8.9	1/11/2020		3.6	7/11/2020	0.094	5/3/2022			0.00209	8/20/2019
	8.6	1/16/2020		3.7	7/14/2020	2	6/1/2022			0.00027	10/22/2019
	7.1	1/24/2020		3.2	7/20/2020	2	6/5/2022			0.00048	1/13/2020
	8	1/28/2020		3.6	7/28/2020	2	6/16/2022			0.00051	4/20/2020
	7.9	2/3/2020		5.2	3/24/2021	2	6/20/2022			0.00059	11/9/2020
	8.8	2/9/2020		5.1	4/12/2021	2	6/28/2022	Not Detected		0.00035	1/19/2021
	6.3	2/14/2020		5.9	5/6/2021	2	8/2/2022	9/27/2019 (<4.5 ug/l)		0.00033	4/27/2021
	6.7	2/20/2020		6.2	6/29/2021	2	8/10/2022	10/22/2019 (<4.6 ug/l)		0.00063	7/20/2021
	6.8	2/26/2020		6.7	2/7/2022	2	8/18/2022	10/23/2019 (<10 ug/l)		0.0003	10/25/2021
	7.9	3/3/2020		7.8	2/23/2022	2	8/21/2022	12/5/2019 (<4.6 ug/l)		0.0003	4/27/2022
	9	3/9/2020		5.1	3/11/2022	2	8/26/2022	12/19/2019 (<4.4 ug/l)		0.00282	7/10/2022
	10.2	3/18/2020		5.5	3/19/2022	2	8/29/2022	1/8/2020 (<4.5 ug/l)			
	10	3/21/2020		5.2	3/23/2022			2/19/2020 (<4.5 ug/l)			
	10.6	3/27/2020	0.093	4/11/2022				3/11/2020 (<4.7 ug/l)			
	10.2	4/2/2020	0.11	4/19/2022				4/9/2020 (<4.5 ug/l)			
	10.7	4/8/2020	6.6	6/1/2022				5/5/2020 (<4.5 ug/l)			
	10.8	4/14/2020	5	6/5/2022				5/14/2020 (<4.5 ug/l)			
	8.1	4/19/2020	5	6/16/2022				6/9/2020 (<4.4 ug/l)			
	14.4	4/24/2020	5	6/20/2022	Not Detected			7/13/2020 (<4.4 ug/l)		Not Detected	
	16	4/27/2020	5	6/27/2022	5/9/2018 CSI (<1.0 ug/l)			8/13/2020 (<4.4 ug/l)		2/22/2018	
	11.3	5/7/2020	5	8/2/2022	9/4/2019			9/18/2020 (<4.5 ug/l)		8/30/2018	
	15.5	5/13/2020	5	8/10/2022	9/14/2019			10/14/2020 (<4.5 ug/l)		7/20/2020	
	17.5	5/18/2020	5	8/18/2022	10/2/2019			11/9/2020 (<4.4 ug/l)		1/11/2022	
	11.6	5/23/2020	5	8/21/2022	10/12/2019			12/10/2020 (<4.5 ug/l)			
	13	5/26/2020	5	8/26/2022	11/6/2019			1/12/2021 (<4.6 ug/l)			
	9.4	6/3/2020	5	8/29/2022	11/9/2019			2/11/2021 (<4.7 ug/l)			
	8.4	6/11/2020			11/14/2019			3/1/2021 (<4.6 ug/l)			
	7.7	6/19/2020			12/11/2019			4/7/2021 (<4.4 ug/l)	QL <0.5 ng/l		
	9.2	6/22/2020			12/19/2019			5/11/2021 (<4.6 ug/l)			
	11.4	6/28/2020			1/2/2020			6/10/2021 (<3.9 ug/l)			
	10.9	7/2/2020			1/8/2020			7/13/2021 (<4.0 ug/l)			
	9.6	7/8/2020			1/11/2020			8/10/2021 (<4.4 ug/l)			
	8	7/11/2020			1/16/2020			9/1/2021 (<4.1 ug/l)			
	8.5	7/14/2020	Not Detected		1/24/2020			10/5/2021 (<4.1 ug/l)			
	8.8	7/20/2020	9/14/2019		1/28/2020			11/17/2021 (<4.1 ug/l)			
	10	7/28/2020	10/2/2019		2/3/2020			12/8/2021 (<4.2 ug/l)			
	8.4	8/2/2020	11/9/2019		2/9/2020			1/11/2022 (<4.1 ug/l)			
	8	8/7/2020	11/14/2019		2/14/2020			2/9/2022 (<4.2 ug/l)			
	6.1	8/13/2020	12/11/2019		2/20/2020			3/7/2022 (<4.3 ug/l)			
	6.4	8/19/2020	12/19/2019		2/26/2020			4/5/2022 (<4.2 ug/l)			
	6.8	8/25/2020	1/2/2020		3/3/2020			5/19/2022 (<4.1 ug/l)			
	5.8	9/10/2020	1/8/2020		3/9/2020			7/12/2022 (<4.0 ug/l)			
	5.8	9/12/2020	1/11/2020		3/18/2020			8/9/2022 (<4.2 ug/l)			
	5.2	9/16/2020	1/16/2020		3/21/2020						
	7.7	9/22/2020	1/24/2020		3/27/2020						
	6.4	9/28/2020	1/28/2020		4/2/2020						
	5.8	10/11/2020	2/3/2020		4/8/2020						

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Copper	Source	Nickel	Source	Selenium	Source	Bis(2-ethylhexyl)phthalate #	Source	Mercury @	Source
	5.3	10/15/2020		2/9/2020		4/14/2020		QL <## ug/l		
	5.7	10/21/2020		2/14/2020		4/19/2020				
	5.3	11/23/2020		2/20/2020		4/24/2020				
	8.1	12/1/2020		2/26/2020		4/27/2020				
	8.5	12/7/2020		3/3/2020		5/7/2020				
	6.8	12/18/2020		3/9/2020		5/13/2020				
	6.7	12/20/2020		3/18/2020		5/18/2020				
	9.1	12/23/2020		3/21/2020		5/23/2020				
	17.5	12/28/2020		3/27/2020		5/26/2020				
	7.4	1/7/2021		4/2/2020		6/3/2020				
	6.5	1/11/2021		4/8/2020		6/11/2020				
	6.1	1/16/2021		4/14/2020		6/19/2020				
	7	1/19/2021		4/19/2020		6/22/2020				
	5.1	2/8/2021		4/24/2020		6/28/2020				
	6.4	2/16/2021		4/27/2020		8/2/2020				
	5.5	3/4/2021		5/7/2020		8/7/2020				
	6.7	3/12/2021		5/13/2020		8/13/2020				
	7	3/15/2021		5/18/2020		8/19/2020				
	7.6	3/20/2021		5/23/2020		8/25/2020				
	8.6	3/24/2021		5/26/2020		9/4/2020				
	7.9	4/1/2021		6/3/2020		9/10/2020				
	8.7	4/9/2021		6/11/2020		9/12/2020				
	9.3	4/12/2021		6/19/2020		9/16/2020				
	6.1	4/18/2021		6/22/2020		9/22/2020				
	7.6	4/21/2021		6/28/2020		9/28/2020				
	15.5	4/27/2021		8/2/2020		10/11/2020				
	6.6	5/6/2021		8/7/2020		10/15/2020				
	5.7	5/8/2021		8/13/2020		11/4/2020				
	7	5/14/2021		8/19/2020		11/13/2020				
	7.2	5/17/2021		8/25/2020		11/19/2020				
	11.6	5/25/2021		9/4/2020		11/21/2020				
	5.38	6/2/2021		9/10/2020		11/23/2020				
	5.11	6/18/2021		9/12/2020		12/1/2020				
	7.66	6/29/2021		9/16/2020		12/7/2020				
	7.2	7/7/2021		9/22/2020		12/18/2020				
	6.5	7/10/2021		9/28/2020		12/20/2020				
	5.8	7/15/2021		10/11/2020		12/23/2020				
	21.2	8/3/2021		10/15/2020		12/28/2020				
	20.4	8/11/2021		11/4/2020		1/7/2021				
	20.5	8/19/2021		11/13/2020		1/11/2021				
	19.6	8/23/2021		11/19/2020		1/16/2021				
	18.2	8/29/2021		11/21/2020		1/19/2021				
	10.3	9/7/2021		11/23/2020		1/27/2021				
	10	9/22/2021		12/1/2020		2/5/2021				
	5.3	9/25/2021		12/7/2020		2/8/2021				
	8.8	9/30/2021		12/18/2020		2/16/2021				
	5.3	10/1/2021		12/20/2020		2/21/2021				
	6.9	10/7/2021		12/23/2020		2/24/2021				
	6.7	10/10/2021		12/28/2020		3/4/2021				
	5.7	10/13/2021		1/7/2021		3/12/2021				
	5.4	11/1/2021		1/11/2021		3/15/2021				
	6.5	11/9/2021		1/16/2021		3/20/2021				
	7.5	11/24/2021		1/19/2021		3/24/2021				
	6.2	12/3/2021		1/27/2021		4/1/2021				
	6.8	12/6/2021		2/5/2021		4/9/2021				
	6.5	12/14/2021		2/8/2021		4/12/2021				
	8.2	12/19/2021		2/16/2021		4/18/2021				

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Copper	Source	Nickel	Source	Selenium	Source	Bis(2-ethylhexyl)phthalate #	Source	Mercury @	Source
	9.6	12/22/2021		2/21/2021		4/21/2021				
	9.4	12/29/2021		2/24/2021		4/27/2021				
	12.5	1/6/2022		3/4/2021		5/6/2021				
	10.1	1/10/2022		3/12/2021		5/8/2021				
	7.7	1/15/2022		3/15/2021		5/14/2021				
	9.9	1/18/2022		3/20/2021		5/17/2021				
	7.6	1/26/2022		4/1/2021		5/25/2021				
	6.6	2/4/2022		4/9/2021		6/2/2021				
	5.7	2/7/2022		4/18/2021		6/6/2021				
	5.7	2/13/2022		4/21/2021		6/10/2021				
	13.9	2/15/2022		4/27/2021		6/18/2021				
	10.1	2/23/2022		5/8/2021		6/21/2021				
	7	3/3/2022		5/14/2021		6/29/2021				
	5.7	3/19/2022		5/17/2021		7/7/2021				
	0.087	4/11/2022		5/25/2021		7/10/2021				
	0.08	4/19/2022		6/2/2021		7/15/2021				
	0.099	4/27/2022		6/6/2021		7/23/2021				
	2.27	5/2/2022		6/10/2021		7/26/2021				
	2.68	5/3/2022		6/18/2021		8/3/2021				
	1.81	5/4/2022		6/21/2021		8/11/2021				
	2.07	5/5/2022		7/7/2021		8/19/2021				
	2.12	5/6/2022		7/10/2021		8/23/2021				
	2.25	5/9/2022		7/15/2021		8/29/2021				
	3.38	5/10/2022		7/23/2021		9/3/2021				
	2.98	5/11/2022		7/26/2021		9/7/2021				
	3.07	5/12/2022		8/3/2021		9/13/2021				
	4.01	5/13/2022		8/11/2021		9/22/2021				
	13.19	5/16/2022		8/19/2021		9/25/2021				
	12.86	5/17/2022		8/23/2021		9/30/2021				
	8.78	5/18/2022		8/29/2021		10/1/2021				
	6.01	5/19/2022		9/3/2021		10/7/2021				
	6.25	5/20/2022		9/7/2021		10/10/2021				
	3.57	5/23/2022		9/13/2021		10/13/2021				
	3.99	5/24/2022		9/22/2021		10/19/2021				
	4.09	5/25/2022		9/25/2021		10/25/2021				
	3.74	5/26/2022		9/30/2021		11/1/2021				
	3.83	5/27/2022		10/1/2021		11/9/2021				
	4.42	5/30/2022		10/7/2021		11/13/2021				
	5.04	5/31/2022		10/10/2021		11/18/2021				
	24	6/1/2022		10/13/2021		11/24/2021				
	5	6/5/2022		10/19/2021		12/3/2021 (QL <5.0 ug/l)				
	5	6/16/2022		10/25/2021		12/6/2021 (QL <5.0 ug/l)				
	5	6/20/2022		11/1/2021		12/14/2021 (QL <5.0 ug/l)				
	5	6/28/2022		11/9/2021		12/19/2021 (QL <5.0 ug/l)				
	5	8/2/2022		11/13/2021		12/22/2021 (QL <5.0 ug/l)				
	5	8/10/2022		11/18/2021		12/29/2021 (QL <5.0 ug/l)				
	5	8/18/2022		11/24/2021		1/6/2022				
	5	8/21/2022		12/3/2021		1/10/2022				
	5	8/26/2022		12/6/2021		1/15/2022				
	5	8/29/2022		12/14/2021		1/18/2022				
				12/19/2021		1/26/2022				
				12/22/2021		2/4/2022				
				12/29/2021		2/7/2022				
				1/6/2022		2/13/2022				
				1/10/2022		2/15/2022				
				1/15/2022		2/23/2022				
				1/18/2022		3/3/2022				
	Not Detected									

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Copper	Source	Nickel	Source	Selenium	Source	Bis(2-ethylhexyl)phthalate #	Source	Mercury @	Source
		9/4/2019 (<10 ug/l)		1/26/2022		3/11/2022				
		9/11/2019 (<10 ug/l)		2/4/2022		3/14/2022				
		9/14/2019 (<10 ug/l)		2/13/2022		3/19/2022				
		9/19/2019 (<10 ug/l)		2/15/2022		3/23/2022				
		9/24/2019 (<10 ug/l)		3/3/2022		3/31/2022				
		10/2/2019 (<10 ug/l)		3/14/2022		5/7/2022 (QL <0.081 ug/l)				
		10/9/2019 (<10 ug/l)		3/31/2022		5/13/2022 (QL <0.087 ug/l)				
		10/12/2019 (<10 ug/l)		4/3/2022 (QL <0.075 ug/l)		5/16/2022 (QL <0.084 ug/l)				
		10/17/2019 (<10 ug/l)		4/8/2022 (QL <0.083 ug/l)		5/24/2022 (QL <0.080 ug/l)				
		10/25/2019 (<10 ug/l)		4/27/2022 (QL <0.092 ug/l)		7/6/2022				
		10/29/2019 (<10 ug/l)		7/6/2022		7/9/2022				
		11/9/2019 (<10 ug/l)		7/9/2022		7/14/2022				
		11/14/2019 (<10 ug/l)		7/14/2022		7/19/2022				
		11/22/2019 (<10 ug/l)		7/19/2022		7/25/2022				
		11/26/2019 (<10 ug/l)		7/25/2022						
		12/4/2019 (<10 ug/l)								
		12/14/2019 (<10 ug/l)								
		12/19/2019 (<10 ug/l)								
		9/4/2020 (<5.0 ug/l)								
		10/9/2020 (<5.0 ug/l)								
		10/27/2020 (<5.0 ug/l)								
		11/4/2020 (<5.0 ug/l)	QL <5.0 ug/l unless otherwise specified			QL <2.0 ug/l unless otherwise specified				
		11/13/2020 (<5.0 ug/l)								
		11/19/2020 (<5.0 ug/l)								
		11/21/2020 (<5.0 ug/l)								
		1/27/2021 (<5.0 ug/l)								
		2/5/2021 (<5.0 ug/l)								
		2/21/2021 (<5.0 ug/l)								
		2/24/2021 (<5.0 ug/l)								
		6/6/2021 (<5.0 ug/l)								
		6/10/2021 (<5.0 ug/l)								
		6/21/2021 (<5.0 ug/l)								
		7/23/2021 (<5.0 ug/l)								
		7/26/2021 (<5.0 ug/l)								
		9/3/2021 (<5.0 ug/l)								
		9/13/2021 (<5.0 ug/l)								
		10/19/2021 (<5.0 ug/l)								
		10/25/2021 (<5.0 ug/l)								
		11/13/2021 (<5.0 ug/l)								
		11/18/2021 (<5.0 ug/l)								
		3/11/2022 (<5.0 ug/l)								
		3/14/2022 (<5.0 ug/l)								
		3/23/2022 (<5.0 ug/l)								
		3/31/2022 (<5.0 ug/l)								
		4/3/2022 (<0.075 ug/l)								
		4/8/2022 (<0.083 ug/l)								
		7/6/2022 (<5.0 ug/l)								
		7/9/2022 (<5.0 ug/l)								
		7/14/2022 (<5.0 ug/l)								
		7/19/2022 (<5.0 ug/l)								
		7/25/2022 (<5.0 ug/l)								

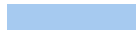
PELSHEET.XLS
5/97

PEQ -- Effluent Data Summary
(all values in ug/L)



Parameter:	3,3'-Dichlorobenzidine #	Source	Hexachlorobenzene # @	Source	Hexachlorobutadiene # @	Source	Hexachlorocyclopentadiene	Source
No. Nondetect	4	MiWaters	4	MiWaters	4	MiWaters	4	MiWaters
Data								
	Not Detected		Not Detected		Not Detected		Not Detected	
	10/23/2019		10/23/2019		10/23/2019		10/23/2019	
	5/5/2020		5/5/2020		5/5/2020		5/5/2020	
	3/8/2021		3/8/2021		3/8/2021		3/8/2021	
	8/16/2021		8/16/2021		8/16/2021		8/16/2021	
	QL <1.5 ug/l		QL <0.01 ug/l		QL <0.01 ug/l		QL <0.01 ug/l	

PEQ -- Effluent Data Summary
(all values in ug/L)



Parameter:	2-Chlorophenol	Source	Bis(2-chloroisopropyl)ether #	Source	Bromomethane	Source	#N/A	Source	#N/A	Source	#N/A	Source
No. Nondetect	4	MiWaters	4	MiWaters	4	MiWaters	0		0		0	
Data												
	Not Detected 10/23/2019 (<10 ug/l) 5/5/2020 (<10 ug/l) 3/8/2021 (<5.0 ug/l) 8/16/2021 (<5.0 ug/l)		Not Detected 10/23/2019 (<10 ug/l) 5/5/2020 (<10 ug/l) 3/8/2021 (<5.0 ug/l) 8/16/2021 (<5.0 ug/l)		Not Detected 10/23/2019 (<1.0 ug/l) 5/5/2020 (<1.0 ug/l) 3/8/2021 (<5.0 ug/l) 8/16/2021 (<5.0 ug/l)							
	(QL <#. # ug/l)		(QL <#. # ug/l)		(QL <#. # ug/l)							

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Copper	Nickel	Selenium	Bis(2-ethylhexylphthalate) #	Mercury @	Perfluorooctane sulfonate @	Perfluorooctanoic acid	Perfluorobutane sulfonate	Cyanide, free	Benzidine #	3,3'-Dichlorobenzidine #	Hexachlorobenzene # @	Hexachlorobutadiene # @	Hexachlorocyclopentadiene	2-Chlorophenol	Bis(2-chloroisopropyl)ether #	Bromomethane	#N/A	#N/A	#N/A
Sample Max.	17.5	7.8	2	4.1	0.0022	0.0078	0.02	0.058	8.4	1.8	0	0	0	0	0	0	0	0	0	0
Total Samples	98	173	173	38	20	8	8	8	28	5	4	4	4	4	4	4	4	0	0	0
Multiplier	0.9	0.9	0.9	1.1	1.4	1.9	1.9	1.9	1.2	2.3	2.6	2.6	2.6	2.6	2.6	2.6	2.6	0	0	0
PEQ:	15.75	7.02	1.8	4.51	0.003948	0.01482	0.038	0.1102	10.08	4.14	0	0	0	0	0	0	0	0	0	0

PROJECTED EFFLUENT QUALITY ANALYSIS (>=10 Detectable):

PARAMETER:	Copper		Nickel		Selenium		Bis(2-ethylhexyl)phthalate #		Mercury @	
TOTAL NO. VALUES:	210		173		173		38		20	
DETECTED:	159		31		17		1		16	
NON-DETECTED:	51		142		156		37		4	
d (% data<detect)	0.242857		0.820809		0.901734		0.973684211		0.2	
m (mean of data >detect)	7.966201		4.671065		1.463176		4.1		0.0008073	
std dev	4.041991		1.595375		0.757198		#DIV/0!		0.0007441	
n	1	30	1	30	1	30	1	30	1	4
d^n	0.242857	3.64E-19	0.820809	0.002675	0.901734	0.044911	0.973684211	0.449308	0.2	0.0016
p	0.933962	0.95	0.720968	0.949866	0.491176	0.947649	-0.9	0.909205	0.9375	0.9499199
std normal dist fn (Z)	2.331321	2.447747	1.597766	2.446652	1.162458	2.428902	#NUM!	2.190504	2.35482	2.4470926
Z_p	1.506269	1.645211	0.585356	1.643912	-0.02205	1.622822	#NUM!	1.336088	1.5344347	1.6444347
1+(s/m)^2	1.257447	1.257447	1.116652	1.116652	1.267809	1.267809	#DIV/0!	#DIV/0!	1.8496649	1.8496649
(sigma_d)^2	0.229084	0.229084	0.110335	0.110335	0.23729	0.23729	#DIV/0!	#DIV/0!	0.6150045	0.6150045
mu_d	1.960666	1.960666	1.486219	1.486219	0.261965	0.261965	#DIV/0!	#DIV/0!	-7.429379	-7.429379
(sigma_dn)^2	0.229084	0.021787	0.110335	0.158069	0.23729	0.288181	#DIV/0!	#DIV/0!	0.6150045	0.2820881
mu_dn	1.960666	1.786111	1.486219	-0.254274	0.261965	-2.037608	#DIV/0!	#DIV/0!	-7.429379	-7.484463
P_95 exponent	2.681607	2.028951	1.680656	0.399311	0.251223	-1.166437	#DIV/0!	#DIV/0!	-6.226042	-6.611072
P_95 (PEQ)	14.60855	7.6061	5.369074	1.490797	1.285597	0.311475	#DIV/0!	#DIV/0!	0.0019773	0.0013454
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

NOTES: For purposes of this summary, ^ represents an exponent or superscript while _ represents a subscript.
Variables are defined in Part 12, Rule 1225 (3) (a).

Perfluorooctane sulfonate @		Perfluorooctanoic acid		Perfluorobutanesulfonate		Cyanide, free		Benzidine #	
8		8		8		28		5	
6		8		8		2		1	
2		0		0		26		4	
0.25		0		0		0.92857143		0.8	
0.003983333		0.0065625		0.0278875		5.6		1.8	
0.002530942		0.00720098		0.017920095		3.95979797		#DIV/0!	
1	30	1	30	1	30	1	30	1	30
0.25	8.674E-19	0	0	0	0	0.92857143	0.10825788	0.8	0.00123794
0.933333333	0.95	0.95	0.95	0.95	0.95	0.3	0.94392998	0.75	0.94993803
2.327251684	2.4477468	2.447746831	2.4477468	2.447746831	2.4477468	0.84460043	2.40048078	1.66510922	2.44724072
1.501385189	1.6452114	1.64521144	1.6452114	1.64521144	1.6452114	-0.51987578	1.58898421	0.67418914	1.64461063
1.40371142	1.4037114	2.204050016	2.20405	1.412915771	1.4129158	1.5	1.5	#DIV/0!	#DIV/0!
0.339119743	0.3391197	0.790296584	0.7902966	0.345655492	0.3456555	0.40546511	0.40546511	#DIV/0!	#DIV/0!
-5.695196161	-5.695196	-5.421531943	-5.421532	-3.752404465	-3.752404	1.52003404	1.52003404	#DIV/0!	#DIV/0!
0.339119743	0.0286398	0.790296584	0.0393505	0.345655492	0.01367	0.40546511	0.39624734	#DIV/0!	#DIV/0!
-5.695196161	-5.827638	-5.421531943	-5.046059	-3.752404465	-3.586412	1.52003404	-0.99983611	#DIV/0!	#DIV/0!
-4.820879681	-5.549214	-3.958961565	-4.719699	-2.785143968	-3.394056	1.1889972	0.00040053	#DIV/0!	#DIV/0!
0.008059694	0.0038905	0.01908292	0.0089179	0.061720203	0.0335722	3.28378659	1.00040061	#DIV/0!	#DIV/0!
Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

Bromomethane		#N/A		#N/A		#N/A	
4		0		0		0	
0		0		0		0	
4		0		0		0	
1		#DIV/0!		#DIV/0!		#DIV/0!	
#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
#DIV/0!		#DIV/0!		#DIV/0!		#DIV/0!	
1	30	1	30	1	30	1	30
1	1	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!	#DIV/0!
Max	Average	Max	Average	Max	Average	Max	Average

REASONABLE POTENTIAL ANALYSIS: WET

Facility: Cadillac WWTP
Outfall: 001

Qe	Qr	RWC	PEL		PEQ*	
			TU _C	TU _A	TU _C	TU _A
4.953560372	0.65	88.40	1.1312188	1	8.40	0.00

* PEQ (enter data for most sensitive species only):

Chronic

Total n =	15	Quant n =	2	CV =	0.6
TU _C effluent =	5.6	MF =	1.5	PEQ TU _C =	8.40

Acute

Total n =	15	Quant n =	0	CV =	0.6
TU _A effluent =	0	MF =	1.5	PEQ TU _A =	0.00

Decision (WET limits/conditions):

WQBEL		Species	TI/RE	Monitoring	
Monthly Avg	Daily Max			Short term	Long term
1.1	1.0	C. dubia & FHM	No		Quarterly

Basis of decision:

Reasonable Potential to exceed Chronic, but not Acute WET.

Recommend a Chronic WET limit of 1.1 Tuc and an Acute WET limit of 1.0 Tua.

Compliance with both test species.

Monitoring frequency of quarterly.

Modified 10/4/00

EGLE
MI0020257 v7.0
Approved
Issued On:11/29/2023
Expires On:10/01/2027

WET Data Summary

Species 1: Ceriodaphnia dubia

Lab	Test Date	TU		Reference
		Acute	Chronic	
	11/7/2017	0.0	0.0	MiWaters
	2/13/2018	0.0	0.0	MiWaters
	5/15/2018	0.0	0.0	MiWaters
	9/6/2018	0.0	0.0	MiWaters
	9/6/2018	0.0	0.0	MiWaters
	11/14/2018	0.0	0.0	MiWaters
	2/12/2019	0.0	0.0	MiWaters
	5/7/2019	0.0	0.0	MiWaters
ERM	8/13-8/19/2019	0.0	1.6	MiWaters
ERM	8/27/2019	0.0	5.6	MiWaters
ERM	10/10-10/17/2019	0.0	0.0	MiWaters
ERM	12/17-12/24/2019	0.0	0.0	MiWaters
ERM	8/17-8/24/2021	0.0	0.0	MiWaters
ERM	3/9-3/15/2021	0.0	0.0	MiWaters
ERM	5/5-5/12/2020	0.0	0.0	MiWaters

Species 2: Fathead minnow

Lab	Test Date	TU		Reference
		Acute	Chronic	
ERM	10/8-10/15/2019	0.0	1.1	MiWaters
ERM	10/17-10/24/2019	0.0	0.0	MiWaters
ERM	5/5-5/12/2020	0.0	0.0	MiWaters
ERM	3/9-3/16/2021	0.0	0.0	MiWaters
ERM	8/17-8/24/2021	0.0	0.0	MiWaters

Modified 10/4/00

EGLE
MI0020257 v7.0
Approved
Issued On:11/29/2023
Expires On:10/01/2027

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Permits Section, WRD, EGLE
Date:	10/17/2022

Outf.	Authorized Flow		Disch. Type *	Receiving Water	Lake?	FLOWS (cfs)			pH Hardness	
	mgd	cfs				95% Exc.	H. Mean	90Q10		
001	3.2	4.95		Clam River	☐	2.6	13.0	3.4	7	70

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

TYPE OF REVIEW: NPDES Permit Reissuance

LOW FLOW FILE NO.: 10194 (Wexford County)

HARDNESS SOURCE: STORET 830054 collected on August 19, 2021 by Permits Section Staff.

PEL SHEET VERSION: 7-22-2022

Review the Following Information Sources

☐ **Permit Application**

☐ **Facility/VGW File**

☐ **MiWaters Documents & Files**

☐ **Current Permit**

☐ **WTA's**

☐ **Staff Report #'s**

☐ **WET Tests**

☐ **Other**

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	70 mg/L CaCO3
Outfall:	001	pH:	7
Date:	10/17/2022	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met. Translator	Background Conc.
		FCV	vd	HNV	vd	HCV	vd	WV	vd	FAV	vd		
Arsenic #	7440382	150	1199707	280	1201009	10	1201009	NA	0	680	1199707	1	1
Antimony	7440360	240	2200106	130	1199807	NA	0	NA	0	2300	2200106	1	0
Barium	7440393	301.4263154	2201912	470000	1202001	NA	0	NA	0	1709.276726	2201912	1	0
Beryllium	7440417	3.656627001	2201401	1200	1201411	NA	0	NA	0	65.81456314	2201401	1	0
Boron	7440428	7200	1202109	330000	1202109	NA	0	NA	0	69000	1202109	1	0
Cadmium	7440439	1.719085414	1199707	130	1199706	NA	0	NA	0	5.793900869	1199707	2.1	0
Chromium	7440473	55.33993022	1199707	9400	1199706	NA	0	NA	0	850.8635019	1199707	1.5	1.6
Lead	7439921	18.56135069	1201801	190	1200709	NA	0	NA	0	181.287016	1201801	4.5	2.5
Molybdenum	7439987	3200	2200604	10000	1200605	NA	0	NA	0	58000	2200604	1	0
Silver	7440224	0.06	1199710	11000	1199705	NA	0	NA	0	1.1	1199710	20	0
Zinc	7440666	87.32624906	1199707	16000	1200510	NA	0	NA	0	173.2354393	1199707	2.1	13
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.6 cfs	Conc.	Load (lbs/day)
Outfall:	001			H. Mean Flow:	13 cfs	Hg Loading Calculated based on LCA (ng/L) of :	0
Date:	10/17/2022			90Q10 Flow:	3.4 cfs	TP Loading based on concentration (mg/L) of :	0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Arsenic #	169.5515938	4.5249929	463.0501563	12.357883	15.90484375	0.42446847	#VALUE!	#VALUE!	680	18.14784	174.8	174.8	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 16 ug/l (0.42 lbs./day), weekly monitoring.	
Antimony	271.4925	7.2455918	215.2921875	5.7457179	#VALUE!	#VALUE!	#VALUE!	#VALUE!	2300	61.3824	3.64	3.64	No Reasonable Potential. No Recommendation.	
Barium	340.9790997	9.1000502	778364.0625	20772.98	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1709.2767	45.617177	52.9	52.9	No Reasonable Potential. No Recommendation.	
Beryllium	4.136445025	0.1103934	1987.3125	53.037396	#VALUE!	#VALUE!	#VALUE!	#VALUE!	65.814563	1.7564591	21.06	21.06	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 4.1 ug/l (0.11 lbs./day), weekly monitoring.	
Boron	8144.775	217.36776	546510.9375	14585.284	#VALUE!	#VALUE!	#VALUE!	#VALUE!	69000	1841.472	624	624	No Reasonable Potential. No Recommendation.	
Cadmium	4.083789472	0.1089882	215.2921875	5.7457179	#VALUE!	#VALUE!	#VALUE!	#VALUE!	12.167192	0.324718	19.95	19.95	Reasonable Potential. Recommend a Mon. Avg. & Daily Max. WQBELs of 4.1 ug/l (0.11 lbs./day) and 12 ug/l (0.32 lbs./day), weekly monitoring.	
Chromium	93.69240004	2.5004628	15566.2315	415.43159	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1276.2953	34.061768	13.8	13.8	No Reasonable Potential. No Recommendation.	
Lead	94.15821878	2.5128945	313.0175781	8.3538131	#VALUE!	#VALUE!	#VALUE!	#VALUE!	815.79157	21.771845	0.174	0.174	No Reasonable Potential. No Recommendation.	
Molybdenum	3619.9	96.607891	16560.9375	441.9783	#VALUE!	#VALUE!	#VALUE!	#VALUE!	58000	1547.904	39	39	No Reasonable Potential. No Recommendation.	
Silver	1.3574625	0.036228	18217.03125	486.17613	#VALUE!	#VALUE!	#VALUE!	#VALUE!	22	0.587136	3.6	3.6	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 1.4 ug/l (0.036 lbs./day), weekly monitoring.	
Zinc	205.7428459	5.4908651	26488.97078	706.93765	#VALUE!	#VALUE!	#VALUE!	#VALUE!	363.79442	9.7089455	94	94	No Reasonable Potential. No Recommendation.	
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

* SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY														
Parameter:	Arsenic #	Source	Antimony	Source	Barium	Source	Beryllium	Source	Boron	Source	Cadmium	Source	Chromium	Source
No. Nondetect	4	MiWaters	2	MiWaters	0	MiWaters	3	MiWaters	0	MiWaters	4	MiWaters	3	MiWaters
Data	76	10/23/2019	1.3	3/7/2021	23	5/9/2018 CSI	8.1	10/23/2019	240	10/23/2019	9.5		1.2	5/9/2018 CSI
			1.4	8/15/2021	15	10/23/2019			170	5/5/2020	0.21		6.9	10/23/2019
					5.3	5/5/2020			170	3/7/2021			0.32	7/11/2020
					8.1	3/7/2021			230	8/15/2021			0.3	7/14/2020
					11	8/15/2021								
		Not Detected		Not Detected				Not Detected				Not Detected		Not Detected
		5/9/2018 CSI		10/23/2019				5/5/2020 (<0.4 ug/l)				5/9/2018 CSI		5/5/2020 (<1.0 ug/l)
		5/5/2020		5/5/2020				3/7/2021 (<1.0 ug/l)				7/14/2020		3/7/2021 (<10 ug/l)
		3/7/2021						8/15/2021 (<1.0 ug/l)				3/7/2021		8/15/2021 (<10 ug/l)
		8/15/2021										8/15/2021		
		QL <1.0 ug/l		QL <1.0 ug/l				QL (<#. # ug/l)				QL <0.2 ug/l		QL (<#. # ug/l)

PELSHEET.XLS
5/97

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Arsenic #	Antimony	Barium	Beryllium	Boron	Cadmium	Chromium	Lead	Molybdenum	Silver	Zinc	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
Sample Max.	76	1.4	23	8.1	240	9.5	6.9	0.087	13	1.8	47	0	0	0	0	0	0	0	0
Total Samples	5	4	5	4	4	6	7	7	3	7	7	0	0	0	0	0	0	0	0
Multiplier	2.3	2.6	2.3	2.6	2.6	2.1	2	2	3	2	2	0	0	0	0	0	0	0	0
PEQ:	174.8	3.64	52.9	21.06	624	19.95	13.8	0.174	39	3.6	94	0	0	0	0	0	0	0	0